

Fit Passport

One body. One fit identity. Any store.

A consumer-owned fit layer for apparel. Project prospectus — 2026.

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This document is a student-project prospectus, not an offer of securities. It describes the problem, the product as built today, the people we build it for, and where it goes next. Figures describing usage are targets and illustrations, not reported results.

1. The problem

Sizes do not agree, and never have. A medium in one brand is a large in another; a 32 waist measures anywhere from 31 to 34 inches of actual cloth; a jacket that fits your shoulders swims at the chest. The label is a guess dressed up as a fact.

The cost of that guess lands on two parties:

- **The shopper** wastes money, time, and trust. Online apparel returns run around a quarter of everything sold, and *"wrong size / wrong fit"* is the single largest reason. Every return is a shipping cost, a refund cycle, and a small erosion of confidence that makes the next purchase harder.
- **Everyone downstream** pays for it — the retailer eats reverse logistics, and the returned garment is often landfilled rather than resold.

The information needed to solve this **already exists** — it is scattered across the one place nobody has organised it: *the clothes that already fit you*. Your closet is a dataset of ground truth about your body, brand by brand, and today it is thrown away at the moment of every new purchase.

Fit Passport turns that closet into a portable profile you own — and translates it to any product page you paste.

2. The product

Fit Passport is a web application. There is nothing to install. It has five parts, and the first four all exist and work today.

2.1 The Passport

Your fit identity as a single object: a metal charge-card — portrait, holder, region, preferred fit, a membership number (No. 00000001), and a verification line. Its **finish is themed by your highest**

earned badge, and you can set it to any metal you've earned. It exports as a PNG and reappears as your banner across the community, so a colour reads as *a person* everywhere in the app. Precise measurements never appear on it.

2.2 The Closet

Your ground-truth wardrobe. Add items by pasting a product URL — the extractor reads brand, garment type, gender, and size chart from the whole link, not just the last path segment. Organise into colour-coded collections in a **filing-cabinet view**: stacked file cards you pull out onto a "desk," a comparison bucket to set items aside, edit history, and honest per-garment fit ratings. This is the dataset the engine learns you from.

2.3 The Size Check

Paste any product link — bare domains are fine. The engine reads the size chart and **ranks every available size, with a per-signal reason for each**. A live multi-region converter sits alongside it. Crucially:

The engine is a transparent rule-and-score model, not a black box and not an LLM. Every recommendation shows its reasoning. An LLM is used only to *extract* product data from a page — it never decides your size. This is a deliberate, load-bearing choice: a fit recommendation you can't interrogate is a fit recommendation you can't trust, and trust is the entire product.

2.4 The Community

The reason to come back. An opt-in directory of members (each wearing their metal banner), an outfit feed with likes, a **follow graph and a followed feed**, and **Ask & Answer** — questions anchored in *real closets*, where an answer can attach a garment you actually own as evidence. Leaderboards surface the day's top looks and the most genuinely helpful members. Everything social is opt-in; everything is moderated (report, block, an admin review queue, rate limits).

2.5 The Badges

The prestige layer that makes contribution worth it — an earned-only ladder of struck-metal medals (see the companion *Badge System* design document). Status here is scarce and cannot be bought, which is precisely what makes people log honest data and help strangers.

3. Why this is defensible

A size calculator alone is a feature, not a company — it can be cloned in a weekend. The moat is not the calculator. It is what the calculator *creates*:

1. **Verified fit context.** Every member has a real closet with brands, sizes, and honest fit ratings. So "this looks good" becomes **"this fits a body like mine, in this size, from this brand."** No general fashion feed can say that, because no general fashion feed knows what actually fits its users.

1. **Earned-only prestige.** Badges and the metal card are derived from real behaviour. Status is visible, scarce, and non-purchasable — a durable reason to contribute that a competitor can't buy their way past.
1. **Crowd fit-knowledge that improves the engine.** "This brand runs small on broad shoulders" is knowledge the community generates and the engine absorbs. Every honest rating and answered question makes the core recommendation better — a compounding, data-driven advantage that widens with scale. **This is the real moat.**
1. **The privacy posture is a feature, not a cost.** Your account code lets someone read your closet and a *coarse* body type — **never your precise measurements**. That constraint is enforced in the data model itself, not by policy, and it's what lets fit be *social* without being invasive.

4. Who we build for

Three overlapping audiences, in priority order:

AUDIENCE	THE JOB THEY HIRE US FOR	WHY THEY STAY
The uncertain buyer	"Will this actually fit me before I pay?"	Fewer returns, more confidence — the entry drug
The taste-builder	"I want to dress better and learn how"	A feed of buyable, wearable looks with provenance
The taste-sharer / blogger	"I want my eye to be seen and to matter"	Earned status; a following built on being right, not loud

The wedge is the **uncertain buyer** — the pain is sharp, frequent, and universal. The *retention* comes from turning that buyer, over time, into a taste-builder and then a taste-sharer. The badge ladder is the rail that carries them along that path.

We deliberately do **not** show scraped brand imagery or logos — brand names appear as text only, and the only photos on the platform are ones users uploaded themselves. This is a considered trademark-and-copyright position, and it doubles as a trust signal: this is *your* data, not a repackaging of the brands'.

5. What is built today

Fit Passport is a working application, not a mockup. As of this writing:

AREA	STATUS
Transparent fit engine with per-signal reasoning	Built
URL extractor (brand / garment / gender / size chart)	Built
Closet with collections, filing-cabinet view, edit history	Built
Passport metal card, badge-themed, PNG export, member numbers	Built
Live multi-region size converter	Built
Outfits, likes, follow graph + followed feed	Built
Ask & Answer with real-closet evidence	Built
Leaderboards (daily top looks, top members)	Built
Badge system — 20 medals, dimensional, turn-to-inspect, WebGL	Built
Anonymous session → claim → shareable account code	Built
Moderation: report, block list, admin review queue, rate limits	Built
Coverage: automated tests across engine, badges, extractor, converters	172 passing

Architecture. Next.js (App Router) · React · TypeScript · Tailwind · Prisma · Zod · Vitest, with Framer Motion + Lenis for motion and a lazily-loaded three.js for the badge inspect view only. Local development runs on SQLite with **no API keys and no cloud services**; production runs on Vercel + Neon Postgres, with the Postgres schema *derived* from the same source so the two can't drift. Sessions are HMAC-signed cookies verified byte-identically on both the Node and Edge runtimes.

Design system. Black-led, on a cool porcelain workspace, with a single cobalt accent spent sparingly, and a characterful editorial serif (Fraunces) paired with a clean sans (Inter) — both self-hosted so a deploy never depends on a font CDN. The aesthetic target is *fashion-forward, high-end, simple, usable* — taste as a first-class requirement, because our users have taste and expect the tools they use to have it too.

6. How it could make money

Not yet monetised — this is a course-stage MVP — but the model is legible and the data asset points at several honest options, in rough order of alignment:

- **Affiliate on the buy.** When we tell you the right size and you buy with confidence, an affiliate link is a natural, non-intrusive cut that aligns us with *the purchase working out* rather than with ad impressions.
- **A fit API for retailers.** The same engine, offered to a store as "will this fit this shopper?", directly attacks their single largest returns cost. Our per-brand fit-truth is the pitch.

- **Premium membership.** Deeper analytics, unlimited closet history, advanced badges and card finishes — paid for by the taste-builders and taste-sharers who get the most from the ecosystem.

Every path is downstream of the same asset: **structured, honest, consumer-owned fit data at scale.** We build that first; monetisation follows the data.

7. Where it goes next

Sequenced smallest-first, so each step proves demand before the next is built:

1. **Deepen the community loop** — richer Ask & Answer, the followed feed as the default home, daily top looks. (*Largely built; now about density.*)
2. **Run one budget styling contest manually** — a \$100 "Thrift Run" — to test whether the fashion audience shows up *before* building event tooling.
3. **Budget contests as a system** — \$100 weekly, \$1,000 monthly, \$10,000 seasonal; itemised-price looks, community voting, commemorative special-metal badges for winners. A budget makes taste comparable and levels the field.
4. **Fit-matched discovery** — filter any feed by "people shaped like me," using the coarse public body type, never measurements. Needs member density to matter.
5. **The retailer fit API** — once the crowd fit-knowledge is deep enough to sell.

The guardrails never move: everything social stays opt-in, precise measurements never become social or matching data, and no step relaxes the privacy invariant no matter how useful it would be.

8. Team & context

Built by **Xiangchen Kong** and **Alyssa Qi** for Carnegie Mellon's *49-800 Start Up Creation in Practice* (Fall 2026), advised by **Prof. Sheryl Root**. The project is a functioning MVP and a live demo, developed under a detailed engineering log, with a Business Model Canvas, Value Proposition Canvas, customer-interview guide, and a risk/legal review maintained alongside the code.

The bet, in one line: **the closet is the dataset, the passport is the interface, and the community is the moat.**

Companion documents: the Badge System design document (this folder); the community ecosystem plan, deployment runbook, and identity/threat model (docs/); the engineering log (DEVLOG.md); and the source itself (app-web/).